

speed. Very low speed could correspond to high quality. Therefore the buyer should not question the resolution but at what resolution does the printer achieve a given speed." Cost and size of acquisition also matter.

Automated bed levelling, multi-material printing and nozzle design are some other factors to keep in mind. Chowdary says, "Metal-body-constructed printer is recommended, and portability is important if repeated movement is required." If you plan to import a printer from outside India, he adds, "packaging for the shipment is critical. It makes a massive difference as 3D printers lose calibration during transport."

Printing material is also of utmost importance. "Buyers prefer the 3D printer to be able to make designs in all materials, and most manufacturers even claim so. But there is a difference between what it can and what it will actually 3D print. Most multi-purpose nozzles get choked with polylactide (PLA), which is a major material used for 3D printing," says Gandhi.

He adds, "So the focus should not only be on the number of different materials that can be used but also how effectively it can 3D print one material without the nozzle getting choked. It took us a while to make an anti-choking nozzle for MY3DBOT for ensuring good customer experience."

Vishal also feels that the user should not be looking at the number of filaments that a printer can use but acceptance of any kind of filament.

Another important parameter, which is not quantitative, is ease of use and after-sales service. Reddy says, "It is important for manufacturers to address any queries or issues faced by the buyer."

Closed versus open printers is another vital parameter to keep in mind. In fact, Vishal feels that support is the first and foremost parameter that a customer should look for before buying a printer. He says, "Service and support is the most vital factor."

Vishal adds, "There must be some kind of assurance in case a problem

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arises. The company must be able to sort it out by sending a technically-qualified person onsite."

Choosing between closed and open is ultimately the buyer's choice. Upputuri notes, "If you are willing to pay a premium for your peace of mind, then go for a closed 3D printer, else an open 3D printer is a better option. As a company in 3D printing space, we have inclination towards open 3D printers."

Bhambri feels customers should focus mostly on parameters that save cost in terms of specifications and accessories. He says, "Reducing downtime, efficiency and ease of printing are other focus parameters."

For schools and universities, do-it-yourself (DIY) 3D printers are apt because of ease of use, learning, and of course, low price point.

Price trends

In the last few years, cost of FDM printers has come down drastically, feel industry experts. Upputuri says, "In the past, printers sold for ₹ 300,000 are now being sold for around ₹ 100,000. In the next one year or so, these printers could even be available for less than ₹ 50,000."

Chowdary too feels that prices of 3D printers are falling as more entrepreneurs are taking the path of manufacturing 3D printers based on open source technologies. He says, "Today, manufacturers are competing on price with very few providing value-added features."

Developments for future

Currently, objects printed using 3D printers are mostly display pieces; these are not functional. In the near

future, you will be able to print a prototype with electronic components like sensors or chips and metallic elements such as screws and ball bearings, informs Reddy.

For instance, he says, "It is possible to print an object up to a certain level, place all electronic and metallic components and resume printing. It is also possible to print up to certain layers, place components at regular intervals and create a fully functional prototype."

He adds, "Once this functionality becomes commercial, it will be a revolutionary change on the FDM side of 3D printing." As of today, Mark One is the only printer that makes fully functional assemblies using carbon-fibre materials.

Mass-automated 3D printing hubs with lowered costs of laser based 3D printing and multi-colour 3D printing using different technologies will be some other important additions in the coming years, feels Chowdary.

"The best technology that we foresee is continuous liquid interface production (CLIP) in which oxidation is used for binding," informs Gandhi.

He adds, "This technology is 20 to 100 times faster than FDM. Instead of making products layer by layer, it makes multiple layers together, increasing the strength of object."

Food printers are slowly entering the market. "Two years down the line, we can foresee food printers emerging in the market," feels Upputuri.

There is also a lot of research on various kinds of materials that can be used to 3D print and on different scanning techniques. He adds, "We also see better technological advancements in digitally acquiring the shape of a 3D object." ●